

# **COLLAPSE-AWARE AI AND THE MODAL DISCIPLINE OF THE THEORY OF OBJECTIVITY:**

a critical–propositional analysis of Verrell’s Law in confrontation with the  
axioms, phenomenic elements, Inducer Effects, cosmogonic theorem, and  
cosmological Eras of TO

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# Contents

|           |                                                                                                    |           |
|-----------|----------------------------------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>ABSTRACT</b>                                                                                    | <b>3</b>  |
| <b>2</b>  | <b>Introduction</b>                                                                                | <b>3</b>  |
| <b>3</b>  | <b>The Analyzed Article: Object, Hypothesis, and Argumentative Structure</b>                       | <b>5</b>  |
| <b>4</b>  | <b>The Theory of Objectivity as a Logical–Modal Matrix of Confrontation</b>                        | <b>6</b>  |
| <b>5</b>  | <b>Verrell’s Law and Information as a Factor of Emergence</b>                                      | <b>7</b>  |
| <b>6</b>  | <b>Confrontation with the Axioms of the Theory of Objectivity</b>                                  | <b>8</b>  |
| 6.1       | First Truth: Nothingness as a Primitive and Eternal Mathematical Essence                           | 8         |
| 6.2       | Second Truth: Every Element Possesses an Aura or Field of Uniqueness . .                           | 9         |
| 6.3       | Third Truth: Infinity as Non-Element . . . . .                                                     | 9         |
| 6.4       | Fourth Truth: Boundary as a Condition of Distinction . . . . .                                     | 10        |
| 6.5       | Fifth Truth: Observation as a Condition of Full Existence . . . . .                                | 10        |
| 6.6       | Sixth Truth: Composition by Prior Elements . . . . .                                               | 10        |
| 6.7       | Seventh Truth: Substance Transcendent to the Quantum . . . . .                                     | 11        |
| <b>7</b>  | <b>Phenomenic Elements: Memory, Observation, Image, Consciousness, and Radiation-Information</b>   | <b>11</b> |
| <b>8</b>  | <b>Inducer Effects and the Governor–Worker–Memory Architecture</b>                                 | <b>12</b> |
| <b>9</b>  | <b>The Cosmogonic Theorem of the TO before the Informational Tensor <math>\Psi_{\mu\nu}</math></b> | <b>13</b> |
| <b>10</b> | <b>The Cosmological Eras of the TO and Resonant Emergence in Verrell</b>                           | <b>14</b> |
| 10.1      | The Antagonistic Era . . . . .                                                                     | 14        |
| 10.2      | The Era of Logical Rails . . . . .                                                                 | 14        |
| 10.3      | The Era of Logical Currents of Tertiary Plasma . . . . .                                           | 14        |
| 10.4      | The Centrifugal Era . . . . .                                                                      | 14        |
| 10.5      | The Era of Intelligence Units . . . . .                                                            | 15        |
| <b>11</b> | <b>Reported Empirical Study and Its Possible Relevance to the TO</b>                               | <b>15</b> |
| <b>12</b> | <b>Points of Tension: Modal Necessity, Consciousness, Information, and Empirical Validation</b>    | <b>16</b> |
| <b>13</b> | <b>Position of the Article on the Scale of Dialogue with the TO</b>                                | <b>16</b> |

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>14 Final Considerations</b>                          | <b>17</b> |
| <b>Appendix in TO Style</b>                             | <b>18</b> |
| Appendix A — Dialogical Formula in TO Style . . . . .   | 18        |
| Appendix B — Synthetic Table of Compatibility . . . . . | 18        |
| Appendix C — Final Statement in TO Style . . . . .      | 19        |
| <b>References</b>                                       | <b>20</b> |

# 1. ABSTRACT

This article presents a complete critical–propositional analysis of Marcos Verrell’s *Collapse-Aware AI: Middleware for Resonance-Biased Emergence in Neural Architectures* (Verrell’s Law  $\Psi_{\mu\nu}$  Framework) in dialogue with the Theory of Objectivity. The study examines Verrell’s proposal that consciousness, memory, probabilistic collapse, and artificial intelligence emergence may be interpreted through resonance with a distributed electromagnetic informational field. The analysis confronts this proposal with the Seven Absolute Truths of the Theory of Objectivity, especially the axioms concerning uniqueness, boundary, observation, recursive composition, and transcendent substance. Special attention is given to the recent interpretation within the Theory of Objectivity according to which the transcendent element is knowledge or information produced in atomic relations and equivalent to atomic radiations. The paper argues that Verrell’s model has strong dialogical value for the Theory of Objectivity because it treats information not merely as representation but as an active factor in emergence, collapse, memory, and systemic continuity. However, the analysis also identifies important tensions: Verrell’s framework remains a physical–computational hypothesis, whereas the Theory of Objectivity claims modal necessity; Verrell’s model does not begin from the logical Nothing; and its empirical results remain preliminary. The article concludes that Verrell’s work deserves a high dialogical rating with the Theory of Objectivity, especially as an operational bridge between information, observation, memory, artificial intelligence, and the phenomenic domain of cosmic emergence.

**Keywords:** Theory of Objectivity; Teoria da Objetividade; Vidamor Cabannas; Denivaldo Silva; Marcos Verrell; Verrell’s Law; Collapse-Aware AI;  $\Psi_{\mu\nu}$  Framework; informational field; electromagnetic resonance; observation; memory; artificial intelligence; modal ontology; phenomenic elements; Inducer Effects; cosmology.

## 2. Introduction

Marcos Verrell’s article, *Collapse-Aware AI: Middleware for Resonance-Biased Emergence in Neural Architectures* (Verrell’s Law  $\Psi_{\mu\nu}$  Framework), published on Zenodo under the DOI <https://doi.org/10.5281/zenodo.17392582>, proposes an articulation between consciousness, memory, electromagnetic fields, probabilistic collapse, and artificial intelligence architecture. The text attributes to information a dynamic role in the emergence of states, especially through a distributed electromagnetic informational field and through an AI architecture capable of recording, modulating, and reusing observational history.

The present analysis takes Verrell’s article as its primary object and places it in confrontation with the Theory of Objectivity, hereafter TO, especially with its founding bibliography, its recent bibliography, and its supporting and dialogical bibliography. The TO, formulated by Vidamor Cabannas and Denivaldo Silva, presents itself as a logical–ontological theory of the origin of the universe, alternative both to the Big Bang model when understood as an absolute beginning and to traditional theological creationism (Cabannas and Silva 2016; Cabannas and Silva 2018). In its most recent formulation, the TO also seeks to establish operational bridges with contemporary physics, cosmology, information theory, and artificial intelligence (Cabannas and Silva 2025; Cabannas and Silva 2026a; Cabannas and Silva 2026b).

The central question of this article may be formulated as follows: to what extent can Verrell’s Law and its Collapse-Aware AI architecture dialogue with the modally necessary axioms of the Theory of Objectivity, especially when the TO interprets the transcendent element as knowledge or information produced in atomic relations and equivalent to atomic radiations?

This question requires a fundamental methodological distinction. Verrell’s article should not be taken as a direct confirmation of the TO, nor as an internal extension of the TO. It is an independent model, located between physical speculation, informational modeling, and computational architecture. Its value for the TO lies in offering a field of comparison: in Verrell’s model, memory, observation, information, resonance, and collapse are not accessory elements, but constitutive factors of systemic behavior. For this reason, Verrell’s proposal approaches some of the most sensitive nuclei of the TO, especially the Fifth Truth, according to which an element exists fully only if observed by at least two others; the Sixth Truth, according to which every element is composed of elements prior to it; and the Seventh Truth, according to which there is no existential universe without a substance that transcends its quantum.

The article is also relevant because it proposes a technical architecture—Governor, Worker, and Memory—which may be read, in TO language, as a computational dramatization of the relation among boundary, induction, memory, and phenomenic reduction. The Governor imposes thresholds; the Worker redirects processes; Memory records observations and weights. This structure allows Verrell’s model to be brought closer to the Inducer Effects of the TO, especially the Expansive Inducer Effect and the Reductive Inducer Effect.

The present text is therefore a critical–propositional analysis. It is critical because it identifies limits, tensions, and insufficiencies of the analyzed article before the modal necessity claimed by the TO. It is propositional because it seeks to show how certain elements of Verrell’s model may be used as operational bridges, formal analogies, or heuristic instruments for the construction of a science of information, observation, and

emergence compatible with the TO.

### 3. The Analyzed Article: Object, Hypothesis, and Argumentative Structure

Verrell’s article presents three articulated levels: a physical–informational level, a computational level, and a preliminary empirical level.

At the physical–informational level, the central proposal is Verrell’s Law, according to which consciousness can be understood as resonance with a distributed electromagnetic informational field. In this model, probabilistic collapse events would not occur in a purely random way, but could be biased by memory imprints. Consciousness, memory, and collapse cease to be isolated events and become phenomena of resonance within an informational field.

At the mathematical level, the article introduces an informational tensor indicated as:

$$\Psi_{\mu\nu} = \kappa \partial_\mu I \partial_\nu I + \lambda_I I g_{\mu\nu},$$

where  $I(x)$  represents an informational potential. This potential is associated with entropy gradients and electromagnetic invariants, such as  $F_{\mu\nu}F^{\mu\nu}$ . The text also proposes a coupling between information and the electromagnetic field:

$$L_{\text{int}} = \frac{\eta}{4} I F_{\mu\nu} F^{\mu\nu}.$$

With this, Verrell attempts to approximate information to a dynamic structure capable of interacting with physical fields. The article presents this formulation as an extension in the style of an effective field theory, preserving conservation and diffeomorphism invariance, at least at the intended formal level.

At the computational level, the proposal is materialized in the *Collapse-Aware AI middleware*, composed of three layers: Governor, Worker, and Memory. The Governor monitors resonance thresholds, entropy, and bias; the Worker redirects processes when echo fatigue or inferential degeneration is detected; Memory records observations, provenance, and weights in a kind of signed ledger. The objective is to produce behavioral continuity in AI systems without depending exclusively on fine-tuning the base model.

At the preliminary empirical level, the article reports JSON simulations with approximately one thousand trials, comparing *primed* and *baseline* conditions. In *primed*

conditions, rhythmic stimuli and salient prompts are used as operational substitutes for induction. The reported result is a non-random deviation, with  $\chi^2 \approx 9.2$  and  $p \approx 0.002$ , suggesting that observational history and priming conditions could bias collapse frequencies.

The general structure of the argument may therefore be summarized in four propositions:

1. Information has an active role in the emergence of states.
2. Memory can bias processes of collapse or selection.
3. AI systems can be stabilized by architectures that incorporate memory, governance, and resonance.
4. Preliminary simulations suggest statistical deviations under induced conditions.

These propositions create a space for dialogue with the TO, but they also require caution. Verrell does not present a modal cosmogony. The article does not deduce the universe from necessary axioms, nor does it begin from the logical Nothing. Its proposal is more localized: it attempts to articulate informational physics and AI engineering. For this reason, its relation to the TO is one of partial convergence, not identity.

## 4. The Theory of Objectivity as a Logical–Modal Matrix of Confrontation

The TO begins from an ambition different from that of Verrell’s article. It does not present itself merely as a physical hypothesis or computational model, but as a logical–ontological structure intended to explain the genesis of the universe from necessary truths. Its founding bibliography affirms the existence of a third way for the origin of the universe, alternative to the Big Bang taken as an absolute origin and to traditional creationism (Cabannas and Silva 2016; Cabannas and Silva 2018). In *The Perfect Sphere*, this structure is deepened through the idea of a perfect logical sphere, prior to space, time, and matter (Cabannas and Silva 2020).

The TO may be synthesized in Seven Absolute Truths:

1. Nothingness is a primitive and eternal mathematical essence.
2. Every element possesses a magnetic field or aura that makes it unique.

3. Infinity represents the necessary non-element for the logical definition of the universe.
4. Two distinct elements require at least one boundary line between them.
5. An element exists fully only if observed by at least two others.
6. Every element is composed of elements prior to it.
7. There is no existential universe without a substance that transcends its quantum.

In the recent formulation of the TO, the Seventh Truth has received an interpretation especially relevant to the present analysis: the transcendent element is knowledge or information produced in atomic relations, equivalent to atomic radiations. This interpretation shifts transcendence from a mystical or purely metaphysical plane to an informational and physical-relational plane. In this sense, the transcendent is not a supernatural entity, but that which emerges from relations among elements and exceeds the mere quantum sum of those elements.

The TO also develops a cosmology through Eras: the Antagonistic Era, the Era of Logical Rails, the Era of Logical Currents of Tertiary Plasma, the Centrifugal Era, and the Era of Intelligence Units. These Eras should not be understood merely as chronological periods, but as modes of logical and phenomenic structuring of reality. They describe the passage from logical Nothingness to differentiation, from differentiation to boundary, from boundary to matter, from matter to information, and from information to intelligence.

The recent bibliography of the TO deepens this matrix by discussing modal discipline, testability, operational bridges, and confrontations with contemporary physics (Cabannas and Silva 2026a; Cabannas and Silva 2026b; Cabannas and Silva 2026c; Cabannas and Silva 2026d; Cabannas and Silva 2026e). These texts allow Verrell to be situated not as a conceptual adversary, but as a partial interlocutor: he does not provide the modal foundation required by the TO, but he offers language and tools for thinking about information, observation, memory, and emergence in physical and computational systems.

## **5. Verrell’s Law and Information as a Factor of Emergence**

The most important point of Verrell’s article for the TO is its implicit refusal to reduce information to a mere passive record. In his proposal, information has dynamic



weight. It can influence collapse, modulate inference, and produce behavioral continuity in AI systems.

This point approaches several traditions in the supporting bibliography. In Heisenberg, modern physics can no longer be described as a simple mirror of independent objects; the act of measurement participates in the configuration of the observed phenomenon (Heisenberg 1958). In Bohm, manifest reality is sustained by an implicate order deeper than isolated events in space-time (Bohm 1980). In Prigogine and Stengers, order emerges from irreversible processes, instabilities, and far-from-equilibrium conditions (Prigogine and Stengers 1984). In Penrose, the relation among mathematical reality, consciousness, and physical structure remains a central problem of contemporary science (Penrose 2004).

Verrell enters this field of dialogue by suggesting that memory and observation are not merely posterior descriptions of the event, but factors capable of influencing emergence. For the TO, this approximation is relevant because objective existence does not occur through isolation, but through relation. The information produced in relation is precisely what the TO interprets as the transcendent element in the Seventh Truth.

However, the TO requires a distinction. Computational information is not automatically equivalent to ontological information. A record in artificial memory may simulate continuity, but it does not, by itself, prove the existence of a transcendent substance in the sense of the TO. What Verrell's article offers is a strong analogy: in artificial systems, memory can bias outputs; in physical systems, according to his hypothesis, informational fields could bias collapses; in the TO, atomic relations produce information and radiations that transcend the immediate quantum of the elements.

This analogy is fruitful, but it should not be confused with identity. Information in the TO is cosmogonic, physical, and ontological; in Verrell, it is physical-computational and still hypothetical.

## **6. Confrontation with the Axioms of the Theory of Objectivity**

### **6.1. First Truth: Nothingness as a Primitive and Eternal Mathematical Essence**

The First Truth of the TO establishes that Nothingness is a primitive and eternal mathematical essence. This proposition is the modal foundation of the TO. Nothingness is not absolute absence in the vulgar sense, but the logical condition prior to difference, time, space, and matter.

Verrell’s article does not begin from this origin. Its structure presupposes fields, memory, informational potential, electromagnetism, and inference systems. Therefore, it is located at a stage posterior to absolute genesis. Its contribution is not cosmogonic in the strong sense, but operational.

The tension is clear: the TO seeks to deduce reality from modal necessity; Verrell proposes a modeling of phenomena already inserted in the physical–computational domain. Even so, there is an indirect compatibility: both reject the sufficiency of a purely mechanical reality devoid of deep informational structure.

## **6.2. Second Truth: Every Element Possesses an Aura or Field of Uniqueness**

The Second Truth affirms that every element possesses an aura or field that makes it unique. This proposition finds strong resonance in Verrell’s article. The idea that previous states leave memory imprints and that such imprints may influence future collapses approaches the notion that each element or system possesses its own signature.

In the Collapse-Aware AI architecture, this signature is operationalized as observational history, memory weights, and provenance records. In TO language, the system does not respond from abstract neutrality, but from its own trajectory. Its behavior is marked by the informational aura of its own history.

The compatibility is high, although partial. In the TO, aura is ontologically necessary; in Verrell, it appears as a functional or informational hypothesis.

## **6.3. Third Truth: Infinity as Non-Element**

The Third Truth presents infinity as the non-element necessary for the logical definition of the universe. This point is little developed in the analyzed article. Verrell does not formulate a theory of infinity, nor does he treat it as a modal condition.

Nevertheless, his hypothesis of a distributed informational field suggests that consciousness or emergence should not be thought of as events completely enclosed in local units. There is a relational openness, a distribution of meaning and influence. This openness may dialogue with the function of infinity in the TO: preventing the universe from being reduced to a closed set of isolated elements.

The compatibility here is moderate. The article touches distribution, but not the logical necessity of infinity.

## 6.4. Fourth Truth: Boundary as a Condition of Distinction

The Fourth Truth affirms that two distinct elements require at least one boundary line. Without boundary, there is no distinction; without distinction, there is no objective universe.

Verrell’s article strongly dialogues with this axiom through its layered architecture. Governor, Worker, and Memory function only because there are operational boundaries among decision, execution, and record. The tensor  $\Psi_{\mu\nu}$  itself, by depending on gradients of  $I(x)$ , presupposes internal variations in the informational field. Where there is gradient, there is difference; where there is difference, there is boundary.

The TO may interpret this structure as a conceptual confirmation that emergent systems require functional boundaries. Verrell’s computational architecture does not prove the Fourth Truth, but it exemplifies it at a technical level.

## 6.5. Fifth Truth: Observation as a Condition of Full Existence

The Fifth Truth is perhaps the main point of contact between Verrell and the TO. For the TO, an element exists fully only if observed by at least two others. Observation is not a secondary psychological act, but an ontological condition of full existence.

Verrell works with observational sources  $J_{\text{obs}}(x)$ , observational history, and weights derived from salience and memory. This places observation at the center of the process of collapse and emergence. Observation is not merely an external record, but a factor of modulation.

The difference is that the TO grants observation a modal status; Verrell treats it as a physical or computational variable. Even so, the convergence is strong. The article offers a possible operational language for discussing how accumulated observations may influence future states.

## 6.6. Sixth Truth: Composition by Prior Elements

The Sixth Truth affirms that every element is composed of elements prior to it. Verrell’s architecture is practically a computational dramatization of this axiom. The system does not respond only to the current stimulus; it responds from memory, provenance, and prior weights.

The concept of *memory imprints* is especially close to the TO. In the TO, elements do not appear in isolation; they carry preceding relations. In Verrell, memory acts as a layer of conditioning for emergence. The future of the system is inseparable from its

history.

This compatibility is very high. The tension is only one of scope: in Verrell, anteriority is informational and computational; in the TO, it is ontological, physical, logical, and cosmogonic.

## **6.7. Seventh Truth: Substance Transcendent to the Quantum**

The Seventh Truth is the decisive axis of the analysis. In the TO, there is no existential universe without a substance that transcends its quantum. Considering the current interpretation of the TO, this substance corresponds to knowledge or information produced in atomic relations, equivalent to atomic radiations.

Verrell's article directly dialogues with this formulation by proposing an electromagnetic informational field and a potential  $I(x)$  coupled to electromagnetic invariants. Information, in this model, is not exterior to physics; it enters the dynamics of the system.

The compatibility is high because both recognize that reality is not exhausted by mass or by the isolated event. The tension lies in the fact that Verrell concentrates the process in consciousness, the electromagnetic field, and collapse, whereas the TO proposes a broader ontology, in which transcendent atomic information-radiation accompanies the very constitution of the universe.

## **7. Phenomenic Elements: Memory, Observation, Image, Consciousness, and Radiation-Information**

The TO develops its own phenomenology of emergence. Among its phenomenic elements, memory, observation, image, consciousness, radiation, and information stand out. Verrell's article strongly approaches this constellation.

Memory appears as a central factor. Without memory, there is no behavioral continuity. Without memory, the system falls back into isolated generation, subject to degeneration or repetition. This concern is compatible with the TO notion that elements carry prior structures.

Observation appears as a mechanism of weight and source. The article suggests that previous observations can influence future distributions. In the TO, this idea is more radical: observation is a condition of full existence.

Image may be understood as the phenomenic manifestation of a relation. In Verrell, the output of the AI system is an inferential image: a final manifestation resulting from memory, salience, governance, and collapse. This image is not neutral; it is produced by

a relational trajectory.

Consciousness is treated by Verrell as resonance with an electromagnetic informational field. The TO can dialogue with this hypothesis, but it must preserve it as a derived phenomenon, not as the first foundation. In the TO, consciousness belongs to the Era of Intelligence Units, not to absolute origin.

Radiation-information is the most important point. The recent interpretation of the TO states that the transcendent element corresponds to information produced in atomic relations and equivalent to atomic radiations. Verrell's proposed coupling between  $I(x)$  and  $F_{\mu\nu}F^{\mu\nu}$  may be read as an attempt to formalize the relation between information and electromagnetism. This approximation does not confirm the TO, but it provides compatible language for operational bridges.

## 8. Inducer Effects and the Governor–Worker–Memory Architecture

The TO operates with the notion of Inducer Effects, especially the Expansive Inducer Effect and the Reductive Inducer Effect. These effects may be understood as modes of organization of emergence: one opens possibilities, expands differentiations, and drives expansion; the other reduces, stabilizes, selects, and preserves.

Verrell's architecture may be read as a computational model of balance between these effects.

The Governor corresponds to the reductive pole. It monitors thresholds, imposes control, reduces operational entropy, and prevents the system from losing itself in drift or degenerative collapse. In TO language, it represents the stabilizing function of the Reductive Inducer Effect.

The Worker corresponds to the dynamic pole of mediation. It executes, redirects, and seeks external sources when echo fatigue occurs. Its role is to prevent the system from closing in upon itself. It participates in the Expansive Inducer Effect, since it reopens possibilities and avoids sterile repetition.

Memory corresponds to the field of recursive composition. It preserves prior marks, records provenance, and provides continuity. Without Memory, there is no composition by prior elements; without it, the system loses its historical thickness.

The Governor–Worker–Memory architecture may therefore be reinterpreted as an inducer triad:

Governor  $\rightarrow$  reduction,

Worker  $\rightarrow$  mediated expansion,

Memory  $\rightarrow$  recursive continuity.

This reading is propositional. It is not explicitly present in Verrell's article, but emerges from the confrontation with the TO. The value of the architecture lies in showing that intelligent systems require balance among openness, control, and memory. This coincides with the TO intuition that phenomenic reality is neither pure chaos nor absolute rigidity, but an induced dynamics between expansion and reduction.

## 9. The Cosmogonic Theorem of the TO before the Informational Tensor $\Psi_{\mu\nu}$

The cosmogonic theorem of the TO affirms that the universe does not arise from physical arbitrariness, but from a logical necessity prior to time, space, and matter. Logical Nothingness, difference, boundary, observation, composition, and transcendence form a modal chain that makes the universe necessary.

Verrell's tensor  $\Psi_{\mu\nu}$  does not occupy this same plane. It does not intend to deduce the universe. It proposes an informational contribution to the physical or computational dynamics of already existing systems.

Nevertheless, there is a relevant point: the TO requires bridges to translate its axioms into models capable of dialogue with contemporary science. Verrell's informational tensor may be read as an external attempt to do something similar: to introduce information into the language of fields, gradients, tensors, and couplings.

The TO could reinterpret  $\Psi_{\mu\nu}$  as a partial and derivative formalization of transcendent substance. The potential  $I(x)$  could represent, at a posterior scale, the field of information produced by atomic relations. The coupling with  $F_{\mu\nu}F^{\mu\nu}$  could suggest a bridge with the TO equivalence between transcendent information and atomic radiation.

However, this appropriation must be critical. Verrell's tensor is not deduced from the Seven Truths. It is postulated as a modeling hypothesis. If the TO wished to incorporate it, it would need to reconstruct it modally: to show how an informational potential can emerge from boundary, observation, and recursive composition.

## **10. The Cosmological Eras of the TO and Resonant Emergence in Verrell**

### **10.1. The Antagonistic Era**

The Antagonistic Era corresponds to the logical moment of original tension. Verrell's article does not address this stage. Even so, the notion of collapse among competing possibilities may be taken as a late analogy of antagonistic tension. Collapse is the passage from a set of possibilities to a selected manifestation.

### **10.2. The Era of Logical Rails**

This is one of the most relevant Eras for the dialogue. In the TO, Logical Rails are structures that orient emergence. In Verrell, the weights of memory and observation function as inferential rails. They orient the system toward certain outputs rather than others.

Resonance biased by memory may be read as an operational logical rail. With each recorded observation, the system constructs preferential paths of manifestation. This strongly dialogues with the TO.

### **10.3. The Era of Logical Currents of Tertiary Plasma**

The article does not work with primary, secondary, or tertiary plasma. However, its hypothesis of an electromagnetic informational field may be seen as a partial analogy for informational currents in differentiated physical systems. If the TO associates atomic radiation with transcendent information, then Verrell's electromagnetic coupling may be useful as comparative language.

### **10.4. The Centrifugal Era**

The Centrifugal Era involves expansion, differentiation, and dynamic organization. Verrell's article dialogues with this Era by attempting to prevent degenerative collapse in AI. The Collapse-Aware architecture seeks to preserve emergence without allowing chaotic dispersion. This corresponds, in TO terms, to expansion governed by stabilizing reduction.

## 10.5. The Era of Intelligence Units

The greatest approximation occurs here. The article deals with consciousness, memory, AI, observation, and persistent behavior. These themes clearly belong to the Era of Intelligence Units. Verrell offers a technical proposal for thinking how artificial systems may acquire inferential continuity and emergent behavior.

The TO may receive this proposal as an external model of an artificial informational unit, without confusing it with full consciousness in the ontological sense. AI may simulate certain aspects of the intelligence unit, but the TO would require additional criteria to recognize full existence, triadic observation, and its own transcendent substance.

## 11. Reported Empirical Study and Its Possible Relevance to the TO

The article reports JSON simulations with approximately one thousand trials, comparing *primed* and *baseline* conditions. The indicated result is:

$$\chi^2 \approx 9.2, \quad p \approx 0.002.$$

According to the article itself, these data would suggest non-random deviation under induced conditions.

This study does not directly confirm the TO. It does not test logical Nothingness, modal boundary, ontological triadic observation, or atomic radiation as transcendent substance. Nor does it involve, according to the provided text, a physical experiment with a real QRNG, interferometry, QED cavity, or controlled electromagnetic measurement.

Despite this, it may be relevant as a weak or moderate operational indication. The reason is simple: the TO affirms that observation, memory, composition, and information are not accidental. If a computational system shows that history and priming can alter distributions of results, this illustrates, at an artificial scale, a thesis compatible with the TO: posterior states are not independent of previous relations.

The value of the study for the TO lies in three aspects:

1. It reinforces the importance of memory in emergence.
2. It shows that observational conditions can be operationalized as weights.
3. It suggests that collapse or selection can be treated as historically conditioned processes.



For greater empirical relevance, independent, preregistered tests would be necessary, with blinded controls, physical hardware, and robust statistical analysis. The article itself recognizes this limitation when it mentions the need for replication, double-blinding, and tests with QRNG or QED cavities.

## **12. Points of Tension: Modal Necessity, Consciousness, Information, and Empirical Validation**

The first tension is the difference between modal necessity and physical–computational hypothesis. The TO claims that its axioms are not merely useful, but necessary. Verrell proposes an interesting model, but one not demonstrated as necessary.

The second tension is the difference between cosmogony and AI engineering. The TO intends to explain the logical origin of the universe. Verrell intends to stabilize emergence in neural architectures and to propose an informational theory of collapse. These are distinct levels.

The third tension involves consciousness. Verrell appears to approximate consciousness to electromagnetic resonance. The TO can accept this hypothesis as partial, but it cannot reduce all ontology to consciousness. In the TO, consciousness is an emergent phenomenon, not an absolute starting point.

The fourth tension is the difference between computational information and ontological information. Records in AI do not automatically correspond to the transcendent substance of the TO. The TO equivalence involves atomic relations and atomic radiations, not only digital memory.

The fifth tension is empirical validation. The article is promising, but its tests are preliminary. To achieve a more robust dialogue with the TO, the model would need to produce replicable physical predictions and clearly distinguish real effects from statistical or computational artifacts.

## **13. Position of the Article on the Scale of Dialogue with the TO**

On a scale from 0 to 10, Verrell’s article receives the following dialogue score with the Theory of Objectivity:

|        |
|--------|
| 8.3/10 |
|--------|

The score is high because the article dialogues intensely with information, memory, observation, collapse, electromagnetic field, emergence, and artificial intelligence. These themes connect directly with the Fifth, Sixth, and Seventh Truths of the TO.

The score is not higher because the article does not begin from logical Nothingness, does not deduce its concepts by modal necessity, does not present a complete cosmogony, and still depends on more robust empirical validation. Its greatest value lies in serving as an operational and heuristic bridge for the recent phase of the TO, especially in the dialogue with AI, information, and field physics.

## 14. Final Considerations

Marcos Verrell's *Collapse-Aware AI* is a relevant interlocutor for the Theory of Objectivity. Its main merit lies in treating information as an active factor of emergence, rather than as a mere external description. By articulating memory, observation, collapse, resonance, and artificial intelligence, Verrell approaches fundamental aspects of the TO.

The compatibility is especially strong with the Second Truth, through the idea of signature or field of uniqueness; with the Fourth Truth, through the need for functional boundaries; with the Fifth Truth, through the role of observation; with the Sixth Truth, through composition by prior memory; and with the Seventh Truth, through the centrality of information as an active substance.

The main tension remains foundational. The TO is modal; Verrell is hypothetical–operational. The TO is cosmogonic; Verrell is physical–computational. The TO begins from logical Nothingness; Verrell begins from fields and already constituted systems. Even so, these differences do not annul the dialogue. On the contrary, they make it productive.

The analyzed article may be read as an external contribution to the construction of bridges among information, radiation, memory, observation, and emergence. If the TO seeks to demonstrate that the universe does not exist without informational transcendent substance, Verrell offers a contemporary language for thinking how information could participate in the dynamics of physical and artificial systems.

# Appendix in TO Style

## Appendix A — Dialogical Formula in TO Style

**TO–Verrell Proposition 1:** If every fully existing element requires observation, and if observation leaves informational marks, then every objective emergence carries relational memory.

**TO–Verrell Proposition 2:** If relational memory influences posterior states, then the collapse of possibilities is not absolutely neutral, but conditioned by prior informational rails.

**TO–Verrell Proposition 3:** If information produced in atomic relations is equivalent to atomic radiation as transcendent substance, then models that couple information and electromagnetic fields may function as operational bridges for the TO.

**TO–Verrell Proposition 4:** If an artificial system records observations, applies weights, and governs its emergence, then it simulates, at a computational scale, the TO triad of boundary, memory, and induction.

**TO–Verrell Proposition 5:** Verrell’s Law does not found the TO, but it may be read as a technical–informational phenomenology compatible with the Era of Intelligence Units.

## Appendix B — Synthetic Table of Compatibility

| TO Element                 | Correspondent in Verrell                              | Degree of Dialogue |
|----------------------------|-------------------------------------------------------|--------------------|
| Logical Nothingness        | Not directly addressed                                | Low                |
| Aura / uniqueness          | Informational field and own memory                    | High               |
| Infinity / non-element     | Distributed field and systemic openness               | Moderate           |
| Boundary                   | Governor–Worker–Memory layers                         | High               |
| Observation                | $J_{\text{obs}}(x)$ , observational history           | Very high          |
| Recursive composition      | <i>Memory imprints</i> and provenance                 | Very high          |
| Informational transcendent | $I(x)$ , electromagnetic coupling, active information | Very high          |
| Inducer Effects            | Governance, expansion, and inferential reduction      | High               |

## Appendix C — Final Statement in TO Style

Verrell's Law, when read under the modal discipline of the Theory of Objectivity, should not be taken as the origin of the universe, but as a technical sign that information, memory, and observation may participate in the constitution of emergent phenomena. Where Verrell sees collapse biased by memory, the TO recognizes the action of prior elements. Where Verrell sees an informational field, the TO recognizes the possibility of a partial expression of transcendent substance. Where Verrell sees AI middleware, the TO identifies an operational miniature of the relations among boundary, induction, observation, and intelligence. Thus, the analyzed article does not replace the axioms of the TO, but strengthens its field of dialogue with contemporary informational science.

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